XRPL Validator — Daily Engineering Report · 2026-09-28

Period: 2026-09-28 00:00 to 24:00 US Eastern (2026-09-28 04:00 to 2026-09-29 04:00 UTC); ledgers #107,284,760 to #107,306,984. Every count from the ledger and the validator's logs stops at #107,306,984. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,207 of the 22,225 ledgers the network closed (99.919%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 18 during planned restarts.
- **Native engine:** 1 engine receipt recorded (1 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 2 deployments (cycles 159, 160), 2 commits merged into main, 1 engine finding fixed (F406).

Service levels

Measure	Value
Ledgers closed by the network	22,225
Validations issued	22,207
Coverage (validations / ledgers closed)	99.919%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	18
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	2
Basis: the hash the first peer announced as accepted (before cycle 183)	22,207

Restarts:

- 14:08:18 UTC: deployment of cycle 159; validations resumed at #107,294,159 (9 ledgers unsigned between the last validation before the restart and the first after it)
- 21:30:57 UTC: deployment of cycle 160; validations resumed at #107,300,981 (9 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,284,760–#107,306,984), by step:

Step	Ledgers with a receipt
Step 1 · native engine	1
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,299,319	Step 1 · native engine	1 transaction result(s) differed from rippled's; 10 ledger change(s) the network made were absent from ours: 29F99D55, D659FE0E, 36BA4DCC; 1 object(s) differed in content	fixed: F406, commit 0145757, deployed in cycle 160 (2026-09-28 21:31 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-28-01, INC-01 is INC-2026-09-28-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
159	`21fdd93`	14:08:18 UTC	n/a	n/a	401 matched, 0 mismatched	#34 from 14:35 UTC: ended early: superseded by the next deployment (2026-09-28 21:31 UTC)
160	`0145757`	21:31:45 UTC	1,693 passed / 0 failed; port 745 / 0	HIST60 60 ledgers, 0 mismatches, WIN98 98 ledgers, 0 mismatches	418 matched, 0 mismatched	#35 from 21:58 UTC: ended early: superseded by the next deployment (2026-09-29 14:58 UTC)

Cycle 159 delivered: `51b0cbe` fix(xrpl-ledger): F405 — an AMM withdrawal by the issuer checks the reserve for its own currency; `21fdd93` fix(xrpl-ffi): link the conan libraries in a fixed order; skip gRPC

Cycle 160 delivered: `0145757` fix(xrpl-ledger): F406 — an AMM equal withdrawal pays the sides in the pool object's order

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
13:14 UTC	xrpl-validator	`21fdd93`	fix(xrpl-ffi): link the conan libraries in a fixed order; skip gRPC
20:13 UTC	xrpl-validator	`0145757`	fix(xrpl-ledger): F406 — an AMM equal withdrawal pays the sides in the pool object's order

Findings fixed

- **fix(xrpl-ledger): F406 — an AMM equal withdrawal pays the sides in the pool object's order** (`0145757`, xrpl-validator) Mainnet #107299319 1E5C6C1A9257: a tfLPToken withdrawal from the XRP/BCFT pool by an account holding 9.381031 XRP at OwnerCount 41 with no BCFT trust line. The transaction names the pool as Asset = BCFT, Asset2 = XRP; the AMM object stores it as Asset = XRP, Asset2 = BCFT. With no Amount or Amount2, AMMWithdraw::applyGuts asks ammHolds for the pool with no assets, which answers in the object's order, so rippled pays …

Other engineering changes

- **fix(xrpl-ffi): link the conan libraries in a fixed order; skip gRPC** (`21fdd93`, xrpl-validator) collect_conan_libs() walked the conan generators directory with fs::read_dir, whose order is the filesystem's (hash order on ext4), so the static-library link order, and with it which archive member supplies a duplicated C++ symbol, differed between hosts. On the build box the differential_probe link failed: a rocksdb object needed a std::string instantiation, the first supplier in that host's order was a gRPC …

Independence program

No independence record for UTC day 2026-09-28; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC
fixBatchV1_2	2026-09-25 14:12 UTC	2026-10-09 14:12 UTC
BatchV1_1	2026-09-25 14:46 UTC	2026-10-09 14:46 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Soak #33: one engine receipt, fixed.** Soak #33 of build ae84a26 ran from 08:28 EDT on 2026-09-27 to 08:32 EDT today and recorded one engine receipt, F405 below. Since that build's deployment at 08:01 EDT on 2026-09-27 the shadow compared 22,690 ledgers and the state-hash check matched 22,701, with no state-hash mismatch: the receipt is a difference in the native engine's own result, found by its comparison against libxrpl, not in the validator's state. The validator stayed on its primary reference node throughout, with no failovers.
- **Finding F405 (AMM withdrawal by the token's issuer).** At ledger 107,278,773 an account withdrew all of its share of an AMM pool of XRP and a token the account itself issues, while holding 1.41 XRP and owning two ledger objects. Before paying out a token, rippled checks that a withdrawer with no trust line for it could afford the reserve for a new one. The check asks only whether that trust line exists, and an issuer never has a trust line to itself, so it applies to the issuer too: a third object needs 1 + 0.2 × 3 = 1.6 XRP, and even with the pool's whole 0.1 XRP the account had at most 1.50999 XRP, so mainnet refused the withdrawal (tecINSUFFICIENT_RESERVE, with the fee charged). The engine exempted issuers from the check, an assumption made when the check was

added on 2026-09-04 that no mainnet transaction had tested until this one, and paid out. The exemption is removed and the mainnet transaction is pinned as a regression vector. Fixed in 51b0cbe on 2026-09-27 and deployed today (below).

- **Deployment.** Build 21fdd93 replaced the live build at 10:09 EDT, after its gate (1,692 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.1 seconds and resumed two ledgers behind the network. The health check matched 401 ledgers with no mismatches, and soak #34 began at 10:35 EDT (seed ledger 107,294,149). Besides F405 it fixes how the build links libxrpl's static libraries: they were linked in the order the filesystem listed them, which differs between machines, so the same link succeeded on one machine and failed on another. They are now linked in a fixed order, and gRPC, which nothing uses, is no longer linked.
- **Soak #34: one engine receipt, fixed.** Soak #34 of build 21fdd93 began at 10:35 EDT and ended about seven hours later with the deployment below. It recorded one engine receipt, F406. Every check during it found no state-hash mismatch (the last at 15:17 EDT); the receipt is a difference in the native engine's own result, found by its comparison against libxrpl, not in the validator's state.
- **Finding F406 (AMM withdrawal: which side is paid first).** At ledger 107,299,319 an account holding 9.381031 XRP and owning 41 ledger objects withdrew its share of an XRP/token pool by redeeming LP tokens, with no trust line for the token. The transaction named the token first and XRP second, but the pool's ledger object stores XRP first, and when a withdrawal names no amounts rippled pays the two sides in the object's order. So the XRP arrived first and counted toward the 9.4 XRP reserve the new trust line needed, and mainnet accepted the withdrawal. The engine paid the sides in the transaction's order, checked the token side against 9.381031 XRP and refused it (tecINSUFFICIENT_RESERVE). It now pays proportional withdrawals in the pool object's order, as its deposit code already did. The mainnet transaction is pinned as a regression vector. Fixed in 0145757.
- **Deployment.** Build 0145757 replaced the live build at 17:31 EDT, after its gate (1,693 suite tests and 745 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches), by warm restart: it rebuilt its state in 22.6 seconds and resumed two ledgers behind the network. The health check matched 418 ledgers with no mismatches, and soak #35 began at 17:58 EDT (seed ledger 107,300,970). It was the first deployment run from the operations team's version-controlled deployment scripts.
- **fixBatchV1_2.** Its source is still unpublished (last checked 21:17 EDT on 2026-09-27); it activates on mainnet about 2026-10-09 14:12 UTC.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Issuer-set TickSize for MPT order books** (dex) Add an optional sfTickSize (UINT8, 3–15) to MPTokenIssuance. It is set at MPTokenIssuanceCreate and can be changed later through MPTokenIssuanceSet on

- **Permissionless batch sweep of expired objects** (fees) Add ttEXPIRED_SWEEP, which carries sfSweepIDs, an STVector256 of at most 16 entries. Any account may submit it. Each keylet must resolve to an object
- **Oracle-banded execution guard for Payment/OfferCreate** (dex) Add an optional sfOracleGuard inner object to Payment and OfferCreate. It holds sfOracleID (the keylet of a g9 AggregateOracle or a single PriceOracle

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-28 13:10	09-28 13:14	xrpl-validator	`21fdd93`	fix(xrpl-ffi): link the conan libraries in a fixed order; skip gRPC
09-28 20:04	09-28 20:13	xrpl-validator	`0145757`	fix(xrpl-ledger): F406 — an AMM equal withdrawal pays the sides in the pool object's order

Times in UTC. A commit appears on the day it was committed to main or merged into it.